

CLINICAL INVESTIGATION

Eye

POSTTREATMENT VISUAL ACUITY IN PATIENTS TREATED WITH
EPISCLERAL PLAQUE THERAPY FOR CHOROIDAL MELANOMAS:
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Purpose: To determine the relationship between the long-term visual function and the dose and dose rates delivered to critical ocular structures in patients with choroidal melanoma treated with ^{125}I episcleral plaque radiotherapy.

Methods and Materials: From 1987 to 1994, 63 patients underwent ^{125}I episcleral plaque (Collaborative Ocular Melanoma Study [COMS] design) application for the treatment of choroidal melanoma. The mean tumor height was 4.5 mm (range 1.7–8.3). Doses and dose rates at the tumor apex, macula, and optic disc were calculated. Forty-three records were scored to assess whether a decrease in visual acuity of >2 lines on a standard Snellen eye chart had occurred. Patient age and the presence of hypertension or diabetes were noted. Statistical analysis was performed to assess both the rate at which visual decline had occurred and the presence of significant factors that had contributed to this decline.

Results: With a median follow-up of 36 months, the 3-year actuarial survival rate was 93.6%. The 3-year actuarial local control rate was 86.9%. The median time to visual loss after therapy was 18.7 months. The 3-year actuarial rate of visual preservation was 40.5%. Multivariate analysis demonstrated higher macula dose rates ($p = 0.003$) to forecast visual decline. Macula dose rates of 111 ± 11.1 cGy/h were associated with a 50% risk of significant visual loss.

Conclusion: Patients in our series treated with ^{125}I plaque brachytherapy for choroidal melanoma experienced favorable tumor control, but with a measurable incidence of visual decline. Higher dose rates to the macula correlated strongly with poorer posttreatment visual outcome. This information may be valuable in selecting the optimal dose rates to treat choroidal melanomas and to predict the risk of visual decline. © 2002 Elsevier Science Inc.

Dose rate, Choroidal melanoma, Visual acuity.

INTRODUCTION

The management of choroidal melanoma, an entity that afflicts approximately 1500 persons in the United States each year, continues to be controversial. Single-institution reports of treatment with ^{125}I episcleral plaque radiotherapy have described high rates of local control and survival comparable to those achieved with enucleation (1–3). A large, prospective randomized trial by the Collaborative Ocular Melanoma Study (COMS) Group has been completed and reported. It showed no significant difference in survival vs. treatment modality for patients with medium-size tumors (4). An equally important question, however, is the functional result achieved by organ-conserving techniques. Currently, published series disagree on the frequency at which pretreatment visual acuity is reduced after plaque application and also differ on the method by which visual decline is measured (2, 5,

6). Broad guidelines for dose prescription have been used in the past in an effort to limit the probability of treatment-related morbidity. Additional analysis of the impact of dose on visual outcome has been addressed by only a few authors (3, 5, 7). This is an increasingly important task, however, as new information becomes available regarding the treatment parameters required for the local control of tumor, which may enable more optimal dose prescription.

In this retrospective review, we investigated the relationship between posttreatment visual acuity and the dose and dose rates delivered to the optic disc and macula. Normal tissue damage at these two sites represents two mechanisms by which visual decline may occur. In addition, we examined the impact of tumor and patient-related factors, including tumor height, tumor distance from the macula and optic disc, age, hypertension, and diabetes mellitus, on vision after therapy.

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Table 1. Patient characteristics

Gender (n)	
Female	20
Male	23
Age (y)	
Median	62.5
Range	23–85
Comorbidities (n)	
Hypertension	19
Diabetes	10

METHODS AND MATERIALS

Patient population and characteristics

From 1987 to 1994, 63 patients underwent ^{125}I episcleral plaque application for the treatment of choroidal melanoma at the Medical College of Wisconsin. Visual acuity data before and after plaque application were available for 52 patients. Nine patients were excluded from analysis secondary to the presence of comorbidities or disease progression that directly affected the visual outcome. The characteristics of the remaining 43 patients and their respective lesions are described in Tables 1 and 2. The diagnosis was made on the basis of the ophthalmoscopic evaluation, ultrasonography, fundus photography, fluorescein angiography, and demonstration of tumor growth patterns over time. A metastatic workup, including liver functions, chest X-ray, and CT scan of the abdomen, was performed for each patient and was negative in all cases.

Treatment methods

The episcleral plaques (Trachsel Dental Studio, Rochester, MN) used were of COMS specification with inner diameters of 12, 14, 16, 18, and 20 mm, carrying 8, 13, 13, 21, and 24 ^{125}I seeds, respectively (8). The seeds were

Table 2. Characteristics of choroidal melanomas

Variable	Mean (mm)	Range (mm)
Height	4.53	1.7–8.3
Diameter	10.64	4.6–16
Distance to macula	2.95	0–12
Distance to optic disc	4.22	0–10

embedded, using a predesigned pattern, in a seed carrier that was then affixed to the plaque using cyanoacrylate. The appropriate plaque size with the corresponding number and activity of ^{125}I seeds was chosen to deliver a dose of 85–100 Gy to the apex at a dose rate of 50–125 cGy/h. The details of plaque placement have been previously published; however, optimal positioning of the seed carrier over the center of the tumor was assumed in treatment planning (9). The ^{125}I seeds were purchased from Medipysics (Arlington Heights, IL).

Basic assumptions were made regarding eye geometry and dimensions (10). These assumptions, the tumor dimensions, and the distances from the tumor to both the macula and the optic disc, as measured by the ophthalmologist, were used to calculate doses and dose rates to these ocular structures and to the tumor apex with the Seed program available on the Theraplan treatment planning system (Kana, Ontario, Canada).

Evaluation of visual acuity and statistical analysis

Visual acuity in the affected eye was measured by the ophthalmologist before treatment and approximately every 4 months after plaque application. The median follow-up time was 36 months. The patient records were scored to assess whether a decrease in visual acuity of at least 2 lines on a standard Snellen eye chart had occurred. The reference

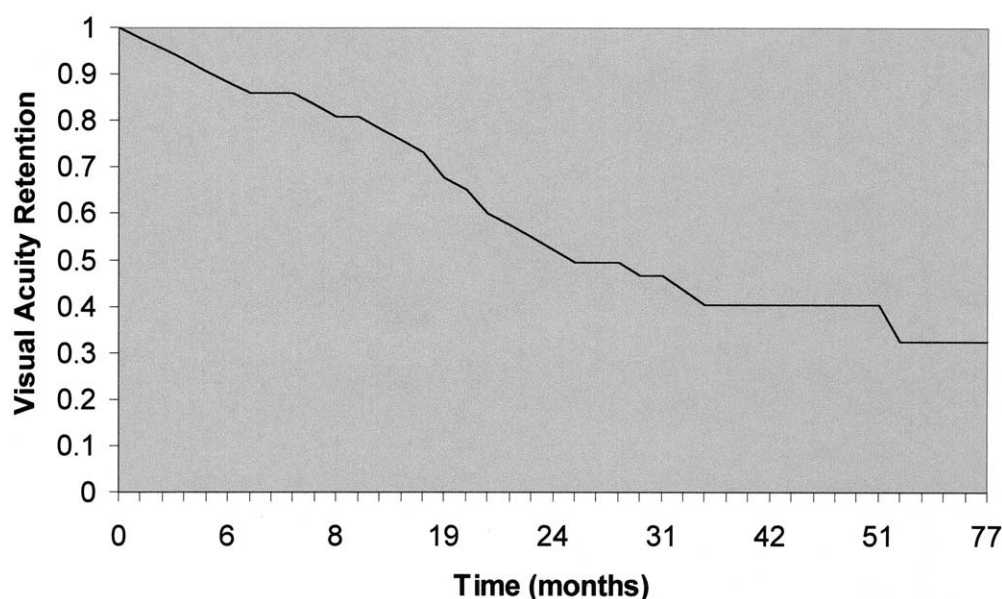


Fig. 1. Fraction of patients ($n = 43$) retaining visual acuity vs. time after plaque application.

Table 3. Dosimetric features of plaque treatment in 43 patients

Variable	Mean	Range
Dose to apex (Gy)	94.3	66.1–132.5
Dose rate to apex (cGy/h)	101.5	58.4–169.9
Dose to macula (Gy)	97.7	9.8–260
Dose rate to macula (cGy/h)	139.2	9–290.2
Dose to optic disc (Gy)	66.5	10.3–200
Dose rate to optic disc (cGy/h)	75.5	9–311.3

visual acuity to which comparisons were made was the best visual acuity achieved before the time of the most recent follow-up examination. A decrease in visual acuity that was clearly secondary to cataract formation alone and that subsequently resolved with therapeutic intervention was not considered to be a “significant visual decline.” The 3-year actuarial local control and overall survival rates were calculated using the Kaplan–Meier method (11). This technique was also used to assess the preservation of visual acuity in the study population. Univariate and multivariate analyses of the variables with an impact on visual acuity were performed using the Cox proportional hazards model (12). Included in the evaluated factors were patient age, macula dose and dose rate, optic disc dose and dose rate, distance of tumor to macula and optic disc, tumor height, and history of hypertension and diabetes mellitus. Logistic regression analysis with inverse prediction was used to derive the 50% tolerance dose (TD₅₀) values for dose rates delivered to ocular structures.

RESULTS

Local control and survival

At the time of analysis, 4 patients had died, 2 without evidence of disease and 2 with metastases. The 3-year actuarial overall survival rate for the study population was 93.6%. Seven patients (16%) experienced local failure. The median time to local failure was 21 months (range 9–28). The actuarial local control rate was 86.9% at 3 years. All 7 patients with local failure eventually required enucleation. No patient underwent enucleation for treatment-related morbidity. Although not the intent of this study, an analysis of patient, tumor, and treatment-related factors that might predict for local control was conducted. On both univariate

and multivariate analyses, none of these factors were found to be significant predictors of local control. Three patients (4.8%) developed metastatic disease at the time of analysis; the site of metastases was the liver in all cases.

Visual acuity: Outcome and determinants

Twenty-five (58%) of the 43 patients analyzed maintained their vision to within 2 lines on the standard Snellen eye chart. The median time to visual loss was 18.7 months. The 3-year actuarial rate of visual preservation was 40.5% (Fig. 1). The etiology of visual loss was not apparent in every case; however, the development of retinopathy was a common event. Twenty-seven patients (62.8%) were found to have some degree of retinopathy during the follow-up period. Although a small subset of these patients improved with treatment, the usual course was one of either no change or further decline of vision.

Table 3 displays the mean dose and dose rates to the macula and optic disc for the study population. Table 4 contains the values of these same variables in the subgroups of patients experiencing visual decline and in those maintaining their visual acuity. Figures 2–5 illustrate the distribution of the values of these variables for all 43 patients. Table 5 details the visual acuity outcome for all 43 patients.

Using proportional hazards univariate analysis, the incidence of visual decline was analyzed as a function of dose and dose rate to the macula and optic disc, distance from tumor to macula and optic disc, tumor height, patient age, past history of hypertension, and past history of diabetes mellitus. The results are shown in Table 6. Increased dose and dose rate to the macula and optic disc and the distance of the tumor to these structures were significant determinants of visual outcome. In contrast, tumor height, patient age, history of hypertension, and history of diabetes mellitus were not significant factors. Multivariate analysis was then performed using the same factors; the proportional hazards model was once again employed. This revealed that an increased macula dose rate ($p = 0.003$) was an independent predictor of decreased visual acuity. Logistic regression analysis with inverse prediction at the 50% probability level was then performed for the macula dose rate. This yielded a value of 111 ± 11.1 cGy/h (95% confidence interval), which predicts for a 50% risk of visual decline.

Table 4. Dosimetric features of patients with and without visual decline after plaque therapy

Variable	NO visual decline		Visual decline	
Dose to macula (Gy)	48.9	(9.8–114.4)	132.88	(121.8–260)
Dose rate to macula (cGy/h)	75.7	(9–280.7)	184.91	(99.6–290.2)
Dose to optic disc (Gy)	36.4	(10.3–107.8)	88.06	(60.6–200)
Dose rate to optic disc (cGy/h)	39.8	(9–186.9)	105.7	(36.7–311.3)
Patient age (y)	66.1	(33–85)	56.44	(23–82 years)
Diabetes mellitus (n)	8		2	
Hypertension (n)	9		10	

Data presented as the mean, with the range in parentheses, except for the last 2 items.

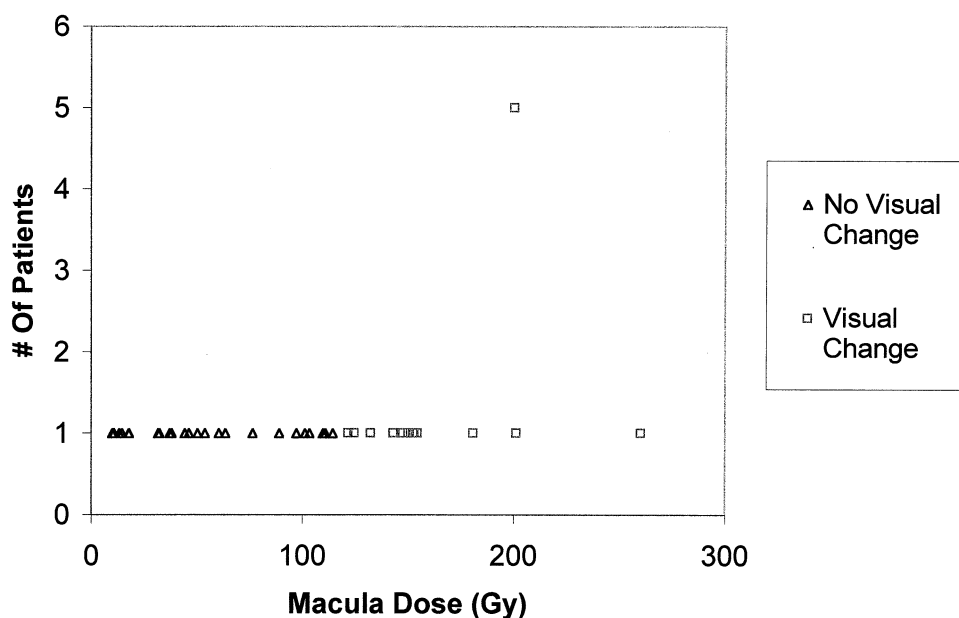


Fig. 2. Dose to macula in patients with and without visual acuity change.

DISCUSSION

Eighty-seven percent of the patients in this review maintained local control actuarially at 3 years, a rate consistent with that described by previous investigators. Quivey *et al.* (3) reported a 5-year actuarial local control rate of 81% in 150 patients treated with ^{125}I plaques at the Wills Eye Hospital. At the University of California, San Francisco (UCSF), the 5-year actuarial rate of local control was 82% in 239 patients treated with ^{125}I (2). Overall survival in patients treated with plaque therapy appears to be equivalent to that achieved with enucleation. Brady and Hernandez (1)

reviewed the results in 302 patients treated with plaque therapy using a variety of isotopes; they reported 5 and 8-year actuarial survival rates of 77% and 67.8% compared with 63% and 43% at 5 and 10 years, respectively, in patients undergoing enucleation. The COMS is attempting to answer, in a randomized, prospective fashion, the question of long-term survival attained with plaque therapy vs. enucleation in patients with medium-size choroidal melanomas.

Organ conservation without preservation of function is not likely to be considered a worthwhile endeavor to most patients. This review found that visual decline secondary to treatment-

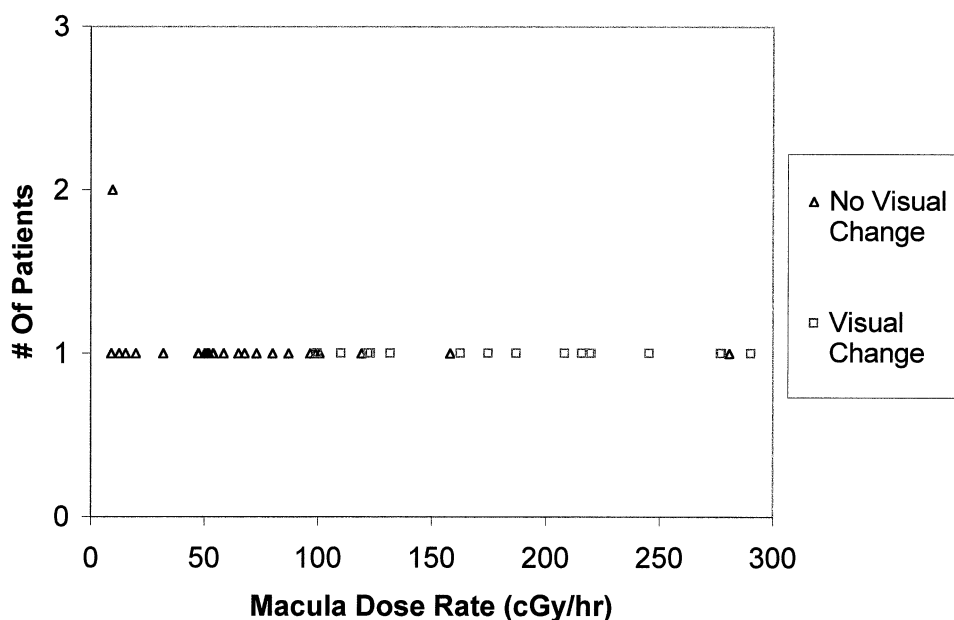


Fig. 3. Dose rate to macula in patients with and without visual acuity change.

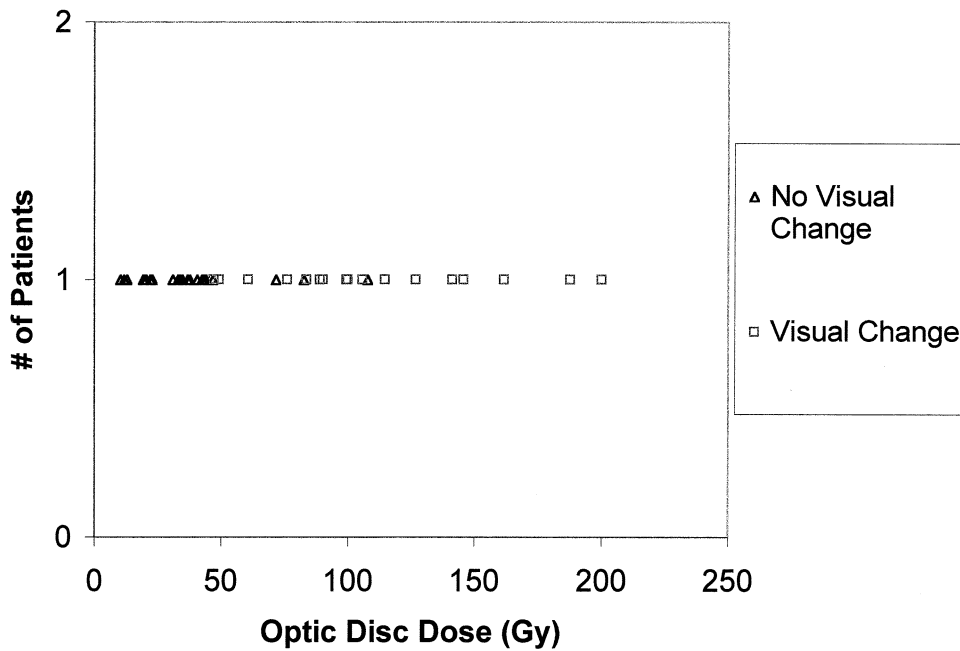


Fig. 4. Dose to optic disc in patients with and without visual acuity change.

related morbidity after ^{125}I plaque therapy for choroidal melanoma is a common event, occurring in 58% of the study population at 3 years. This same conclusion has been reached by several other single-institutional reports, although the rate of visual loss varied somewhat, as did the method of measurement. Several authors reported visual outcome as the percentage of patients maintaining acuity of at least 20/200, generally considered to be a demarcation for useful ambulatory vision. Quivey *et al.* (2) at the UCSF found that 58% of 239 patients treated with ^{125}I plaques maintained their vision at or greater than the 20/200 level. Most recently, the COMS Group re-

ported visual acuity results from their large, multicenter study; 49% of patients experienced a loss of ≥ 6 Snellen chart lines at 3 years (13). A theoretical loss of vision from 20/20 to 20/200 or a loss of 6 lines represents a substantial decrease in visual acuity, however, and thus such reporting methods may underestimate the magnitude of therapy-related changes in vision. For this reason, and because our study population presented with a variety of pretreatment visual acuities, we chose to use a different but common definition of visual loss, a decrease of ≥ 2 lines on the standard Snellen eye chart. Garretson *et al.* (5) at the Mayo Clinic found that 54% of the patients undergoing

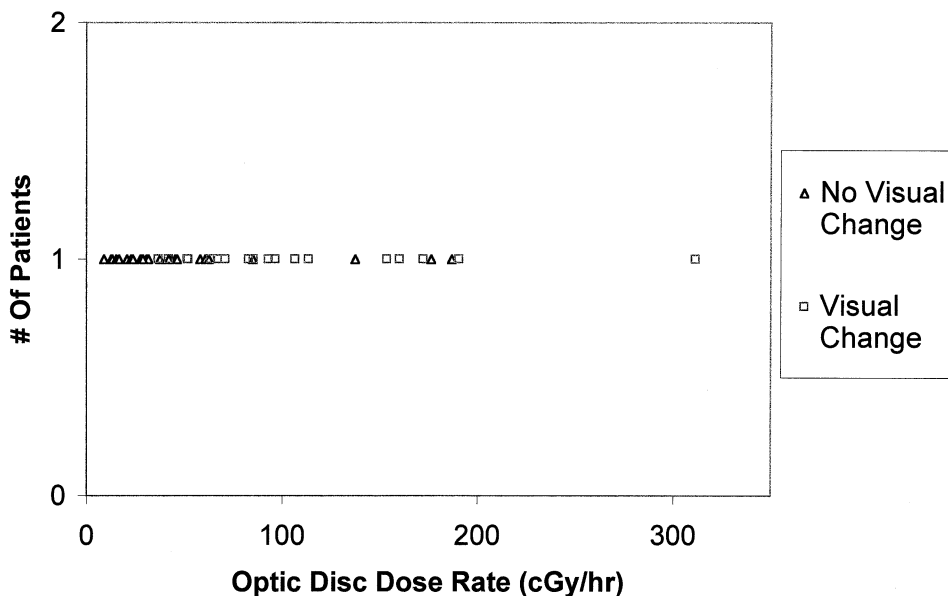


Fig. 5. Dose rate to optic disc in patients with and without visual acuity change.

Table 5. Initial and final visual acuities

Pt. No.	Initial visual acuity (20/×)	Posttreatment acuity (20/×)
1	25	400
2	40	300
3	40	60
4	40	400
5	20	30
6	200	400
7	200	400
8	200	200
9	30	100
10	40	60
11	1000	2000
12	200	400
13	20	40
14	200	200
15	30	40
16	20	40
17	20	25
18	20	70
19	50	60
20	20	30
21	60	200
22	60	80
23	20	200
24	100	200
25	30	40
26	400	400
27	30	100
28	70	300
29	2000	3000
30	25	30
31	20	1000
32	40	100
33	60	200
34	20	60
35	200	200
36	25	40
37	20	80
38	20	30
39	50	400
40	70	400
41	20	200
42	30	200
43	20	30

¹²⁵I brachytherapy maintained their vision to within 2 Snellen lines. Packer *et al.* (6) reported similar results, with a 45% rate of visual preservation to within 2 Snellen lines at 5 years. Our review thus seems consistent with these earlier series.

The causes of visual loss after plaque application are multifactorial. Tumor recurrence or progression jeopardizes sight in two ways: first, by involvement of vital ocular structures and second by requiring additional treatment, most commonly enucleation. This occurred in 7 patients in our series. Cataract formation after plaque therapy is another common mechanism by which visual decline occurs. Fontanesi *et al.* (7) found that 43 of 144 treated patients developed cataract after ¹²⁵I plaque therapy.

Not all cataracts are severe enough to impair vision significantly, however, but for those that do reduce visual acuity,

cataract removal with lens implantation or other corrective measures to reverse the deficit can often be performed. As a result, in the present review, a temporary decrease in visual acuity that was clearly secondary to cataract formation alone was not considered "significant visual decline." Of the remaining patients who had visual decline, however, recovery was the exception rather than the rule. In these cases, multiple abnormalities, including cataracts, radiation retinopathy, neovascular glaucoma, and optic neuropathy, were found in varying degrees on the follow-up examinations. The presence of these changes did not, however, in all cases, predict for visual decline. In fact, 9 patients were noted to experience the same abnormalities with no significant change in visual acuity. This suggests that the normal tissue damage necessary to cause visual decline may not always correlate with the changes seen clinically.

We postulated that the process causing decreased visual acuity, whether singular or multifactorial in nature, is dose related and is most likely to evoke visual decline by acting on either of two critical ocular structures, the macula or optic nerve. This has been examined previously, to an extent, by other authors. Packer *et al.* (6) described radiation retinopathy and subsequent loss of vision as "...invariably due to the proximity of the tumor to the macula," suggesting that the dose given to both tumor and normal structures had an adverse effect. Quivey *et al.* (2) found that both the tumor distance to the optic nerve and fovea and the lesion thickness had an impact on posttreatment visual acuity. The resulting dose to the macula and optic nerve is suggested as the explanation for this phenomenon. Brady and Hernandez (1) reported their results from the Wills Eye Hospital and Hahnemann University for patients treated with a variety of isotopes, including ¹²⁵I. Ninety percent of patients receiving <50 Gy to the fovea maintained a posttreatment visual acuity of 20/200 or better and only 52% of patients with foveal doses >50 Gy retained visual acuity of at least 20/200 (1). Likewise, the results of our review demonstrate that, in univariate analysis, increased doses to the macula and optic nerve, as well as the distances from the plaque to these structures, are significant predictors of visual decline ($p < 0.001$). The COMS Group recently reported that the distance to the foveal avascular zone and tumor thickness were risk factors for poor visual outcome; our study showed no correlation with tumor height or thickness (13). With multivariate analysis, using the proportional hazards model, the effect of the macula and optic disc dose was no longer significant. This is not surprising, because the multivariate analysis also included the tumor distance from these structures, which is inherently related to the dose the macula and optic disc receive. Nevertheless, the dose to the macula and optic disc may be important predictors of visual loss.

Compared with dose, the impact of dose rate on post-treatment morbidity in patients treated for choroidal melanomas with ¹²⁵I applicators is less well understood. Yang *et al.* (14) compared the results in rabbit eyes treated with low-dose-rate ¹²⁵I applicators with those treated with high-dose-rate plaques; the latter resulted in more frequent chorioidal-retinal damage at 24 weeks of follow-up. It has been

Table 6. Cox proportional hazards analysis of treatment variables

Factor	Univariate <i>p</i> value	Risk ratio	Multivariate <i>p</i> value	Risk ratio
Macula dose	<0.0001	1.012	0.0743	1.006
Macula dose rate	<0.0001	1.013	0.0030	1.490
Optic disc dose	<0.0001	1.019	0.7858	1.010
Optic disc dose rate	<0.0001	1.586	0.1850	1.070
Tumor height	0.595	1.040	0.6495	1.009
Distance to macula	<.0001	0.920	0.2590	1.100
Distance to optic disc	<.0001	0.959	0.5404	1.003
Age	0.123	0.991	0.825	.998
Diabetes mellitus	0.907	1.202	0.223	1.585
Hypertension	0.163	0.987	0.153	1.002

unclear whether the same differential in morbidity seen in the report of Yang *et al.* (14) exists over a narrower range of lower dose rates more typically seen in clinical practice. Our investigation found that increased dose rate to the macula was a significant independent predictor of visual decline ($p = 0.003$) in multivariate analysis. This is consistent with the premise that although increased dose rates may result in increased tumor cell kill in melanoma cells secondary to their broad shoulder on the radiation survival curve, an equally, if not more significant, deleterious impact may be encountered in normal tissue toxicity. Indeed, Quivey *et al.* (3) have reported lower dose rates to the tumor apex and base as correlating significantly with a higher rate of local failure. The median dose rate at the tumor base was 98 cGy/h and 76 cGy/h for the nonrecurrent and recurrent groups, respectively. Conversely, Fontanesi *et al.* (9), in an attempt to reduce complications, reduced the apical dose to 75 Gy at a dose rate of 60–65 cGy/h. They described progressive tumor growth in only 4 of 144 patients and retention of vision of 20/200 or better in 59 of 102 patients (9). In a separate report, the same authors reported the results of patients with large lesions treated under the same dosing

guidelines. The tumor distance to the macula and/or optic disc was not described. Thirty-eight of 58 assessable patients had stable or improved posttreatment visual acuity (7). The ideal dose rate for achieving local control without incurring a high risk of decreased visual acuity thus remains undefined.

An analysis of our data revealed a 50% risk of visual decline with a dose rate to the macula of 111 ± 11.1 cGy/h. It is simplistic to believe that this value represents, even within the 95% confidence intervals, concrete definitions of tolerance dose rates. ^{125}I plaque dosimetry is complex, and our methods of dose-rate calculations made assumptions that may not be valid in all cases. In addition, factors not included in our multivariate analysis may contribute significantly to posttreatment visual decline. Despite these caveats, we believe our data suggest that treatment with reduced dose rates, as described by Fontanesi *et al.* (7, 9), may result in both continued local control and a lower rate of visual decline among patients at highest risk of such morbidity. Additional work to define more precisely the optimum treatment parameters will, it is hoped, lead to improved functional outcomes in patients with choroidal melanomas undergoing organ-preservation therapy.

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